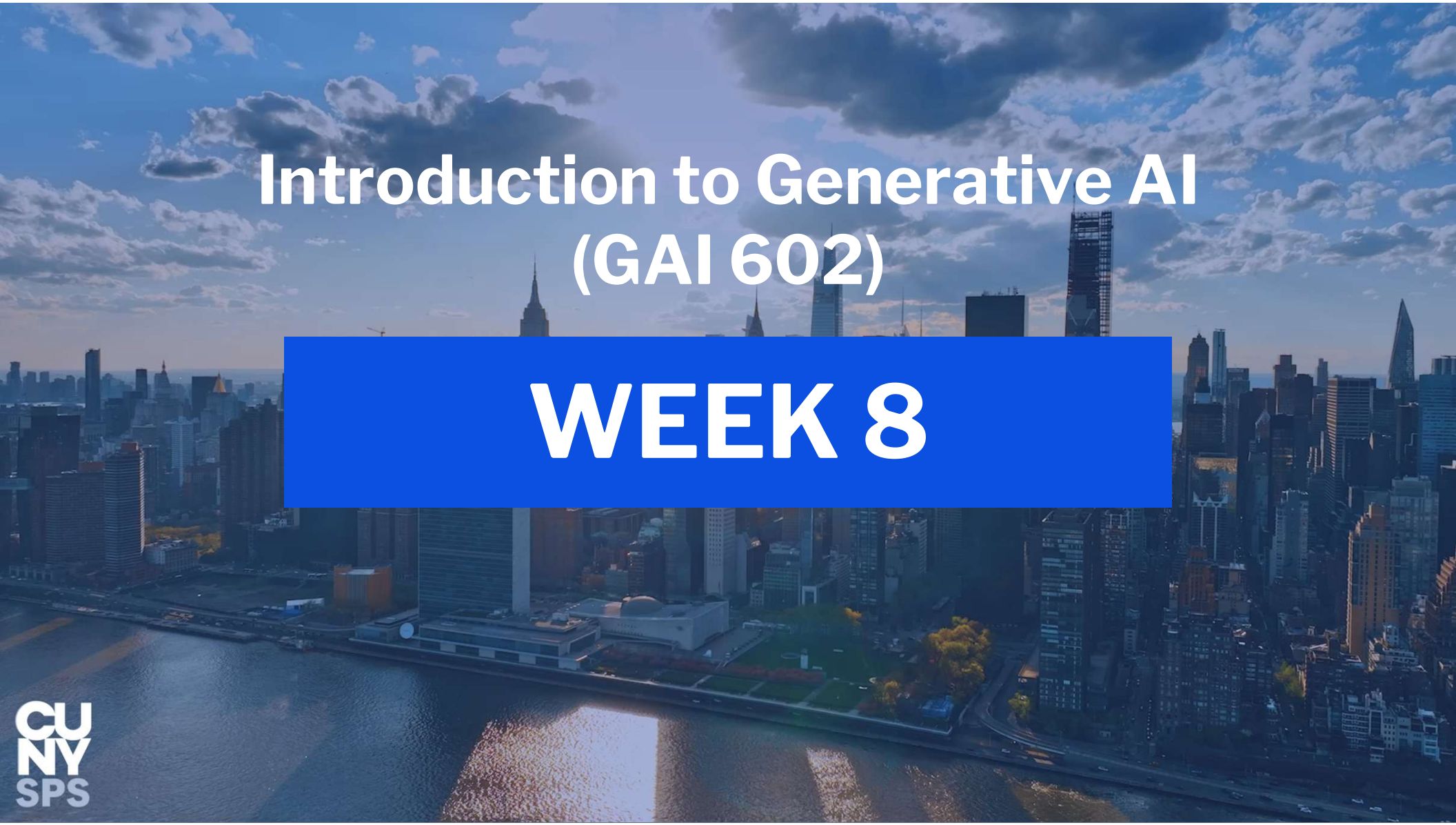
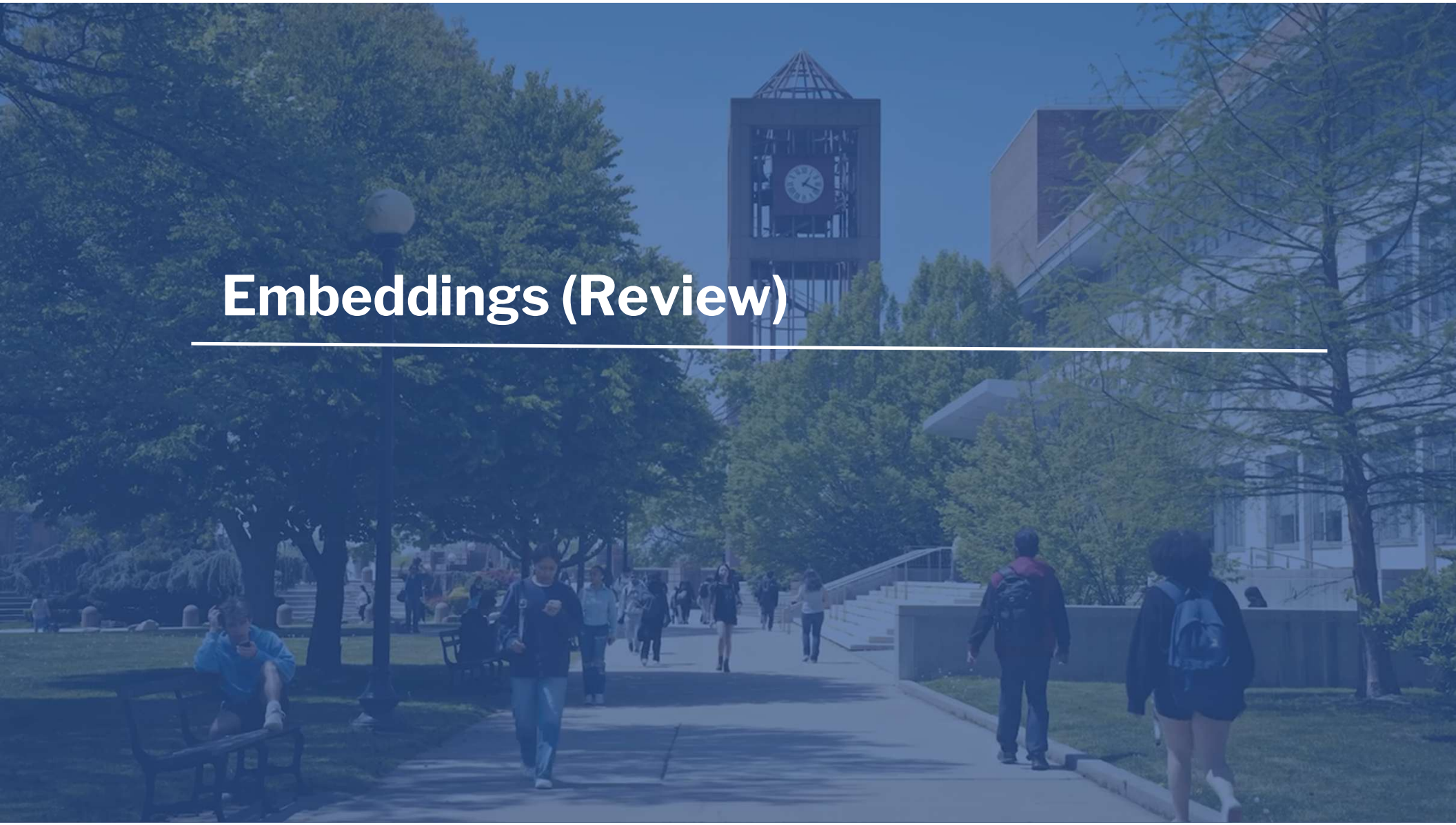
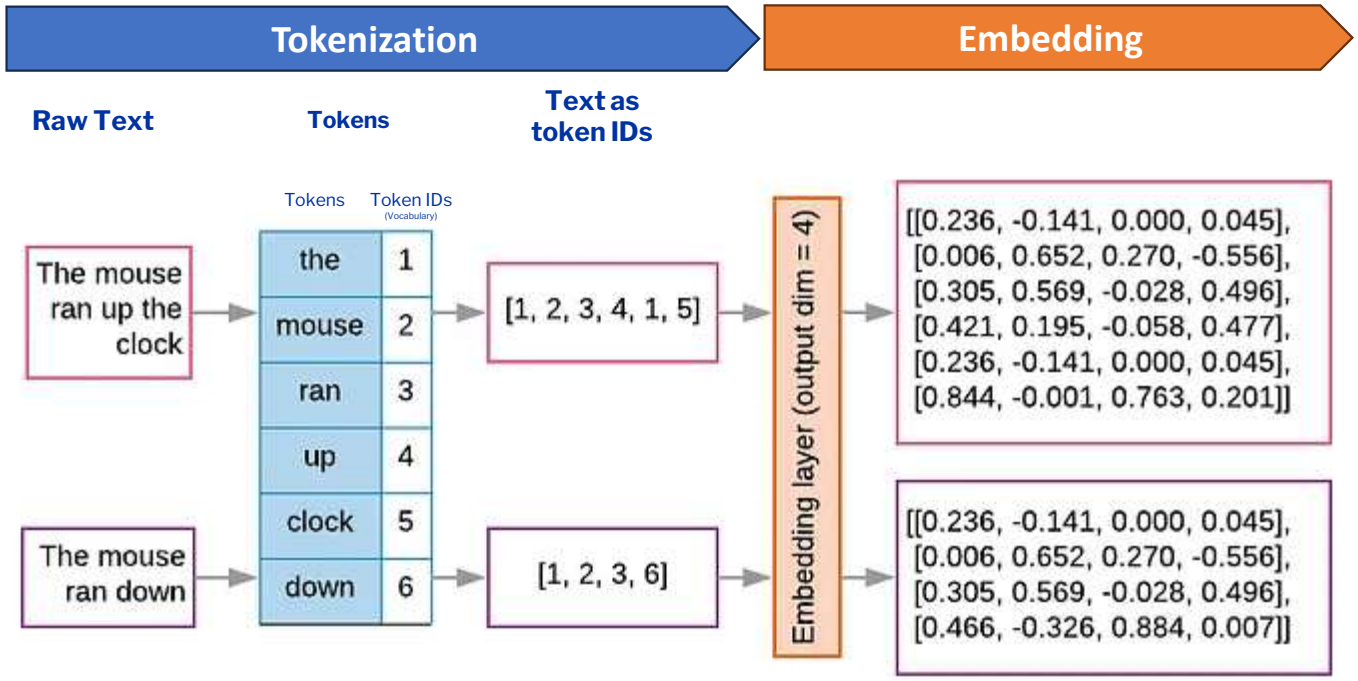

Introduction to Generative AI
(GAI 602)
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Embeddings (Review)

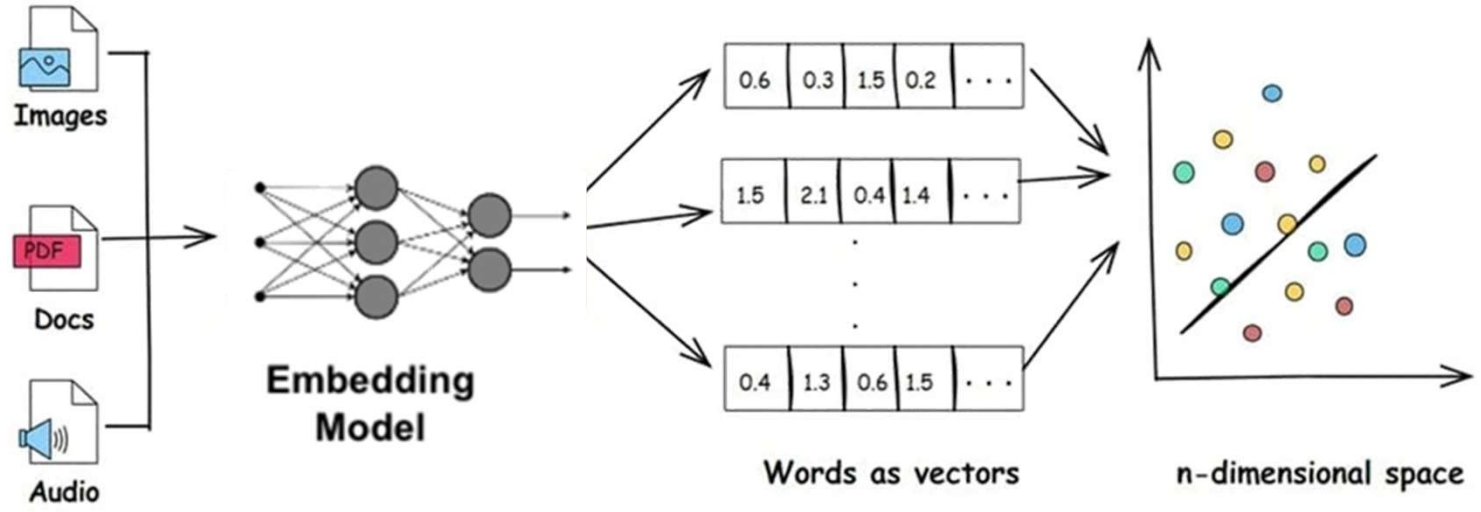
Embeddings

Text embeddings convert words, phrases, or entire documents into dense numerical vectors



Embedding model

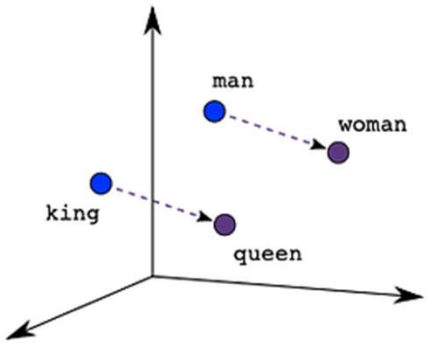
The Embedding Model (transformer-based LLM) generates the embeddings



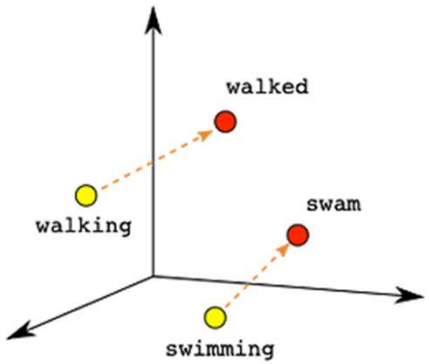
Source: devopscube.com

Embeddings

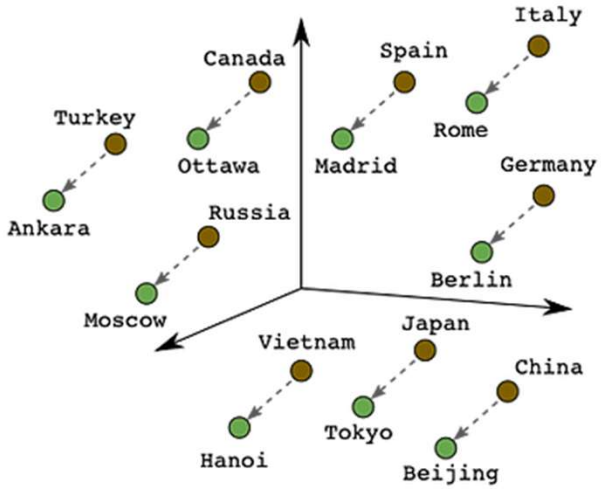
Embeddings should be logically consistent



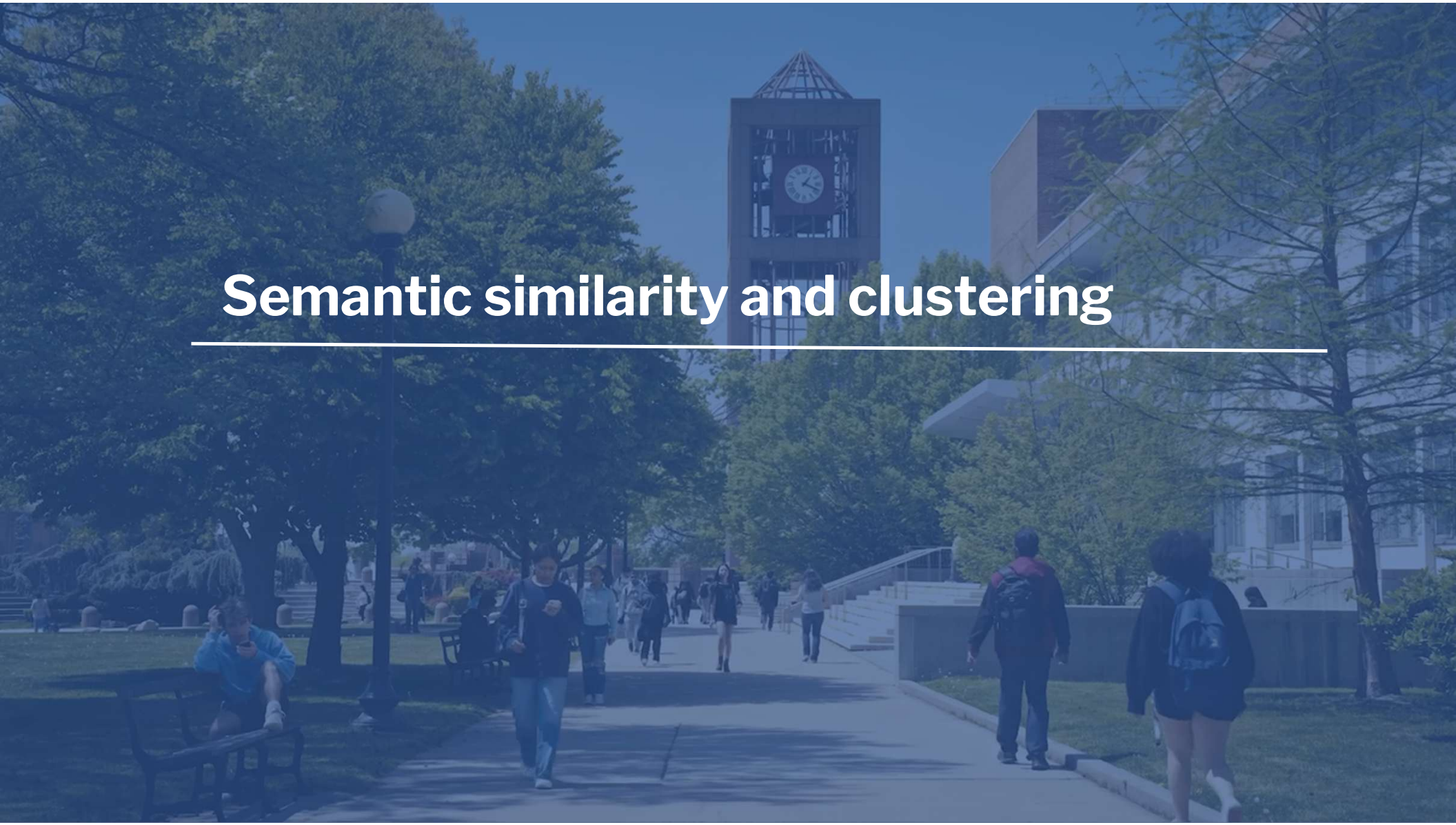
Male-Female



Verb Tense



Country-Capital



Semantic similarity and clustering

Semantic Understanding

Semantic Understanding
- Semantic refers to the meaning i.e. how the text should be interpreted
- Semantics focuses on the meaning of words and how they relate to each other
- Understanding the relationships between words, phrases, and content

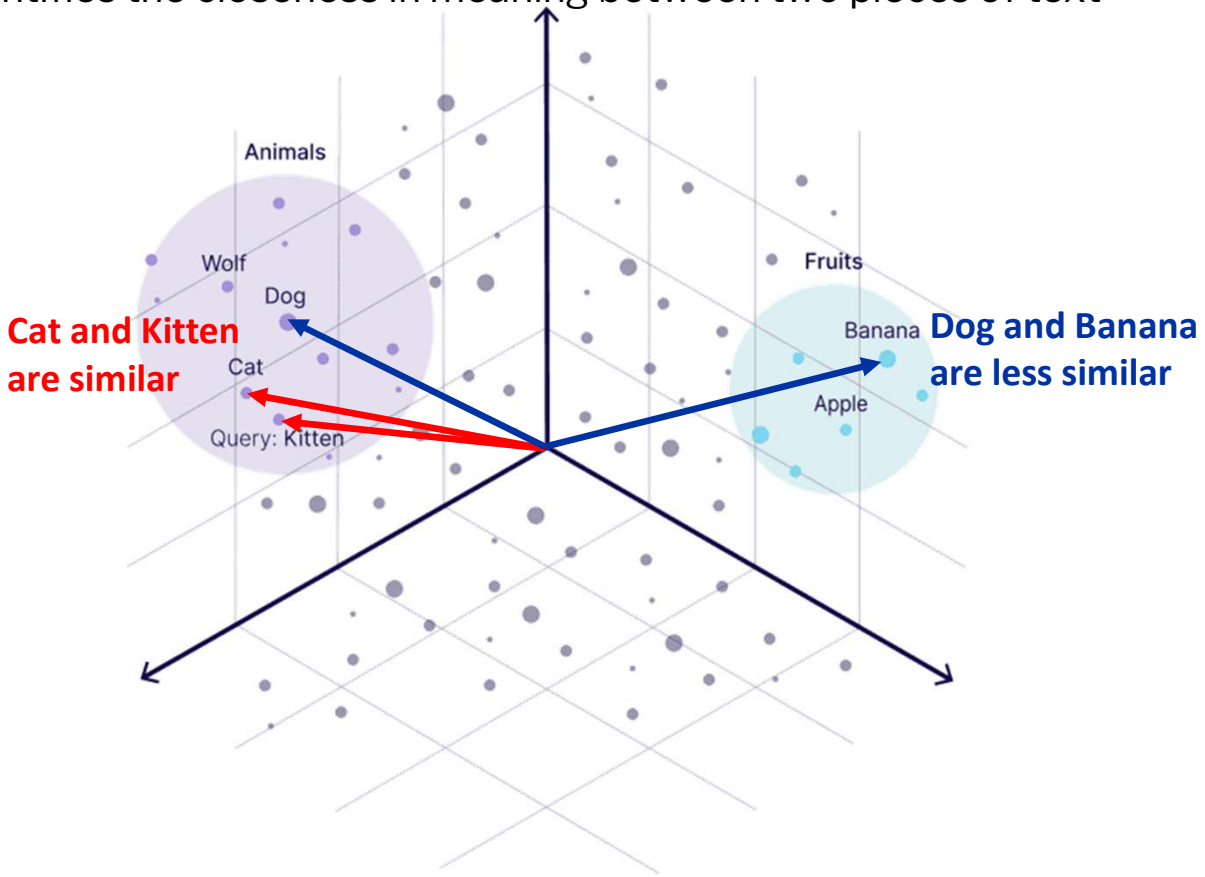


How is Syntax different to Semantics
- Syntax refers to the rules of language e.g. word order, grammar, etc.
- It deals with the structure and order of words, ensuring that they form a valid expression

Source: Google

Semantic Similarity

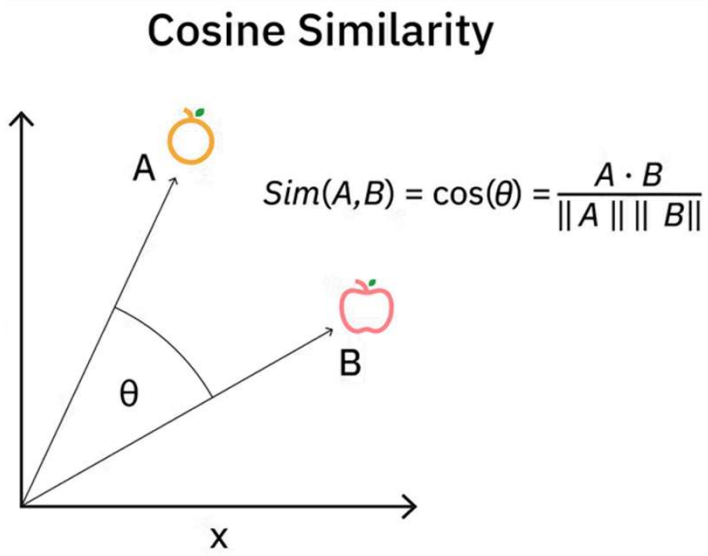
Semantic similarity quantifies the closeness in meaning between two pieces of text

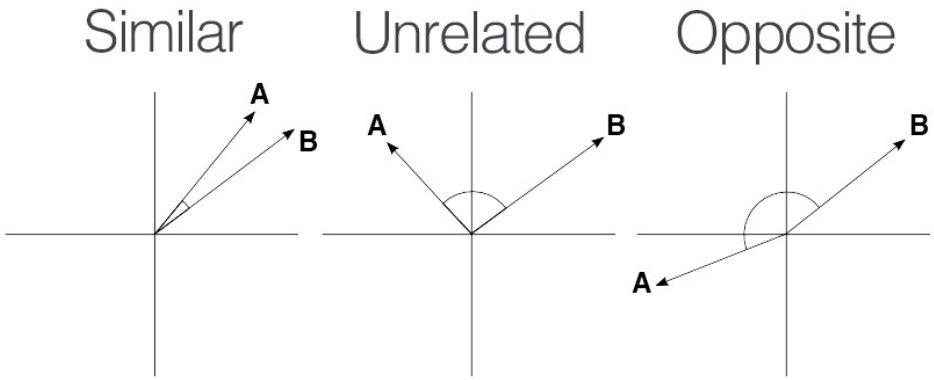


Source: elastic.co

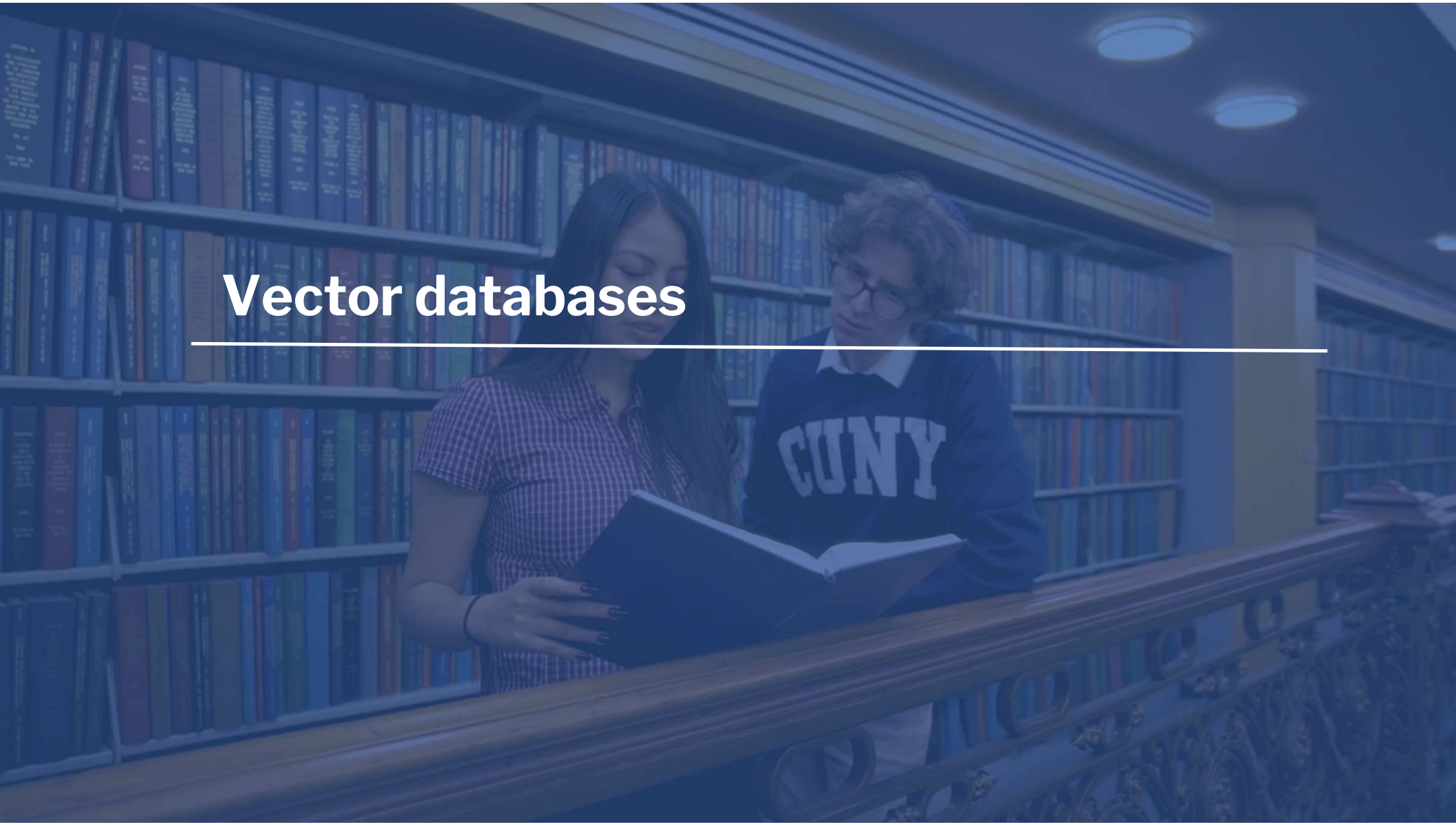
Semantic Similarity

How do we quantify similarity? Cosine similarity is fast and handles high-dimensionality



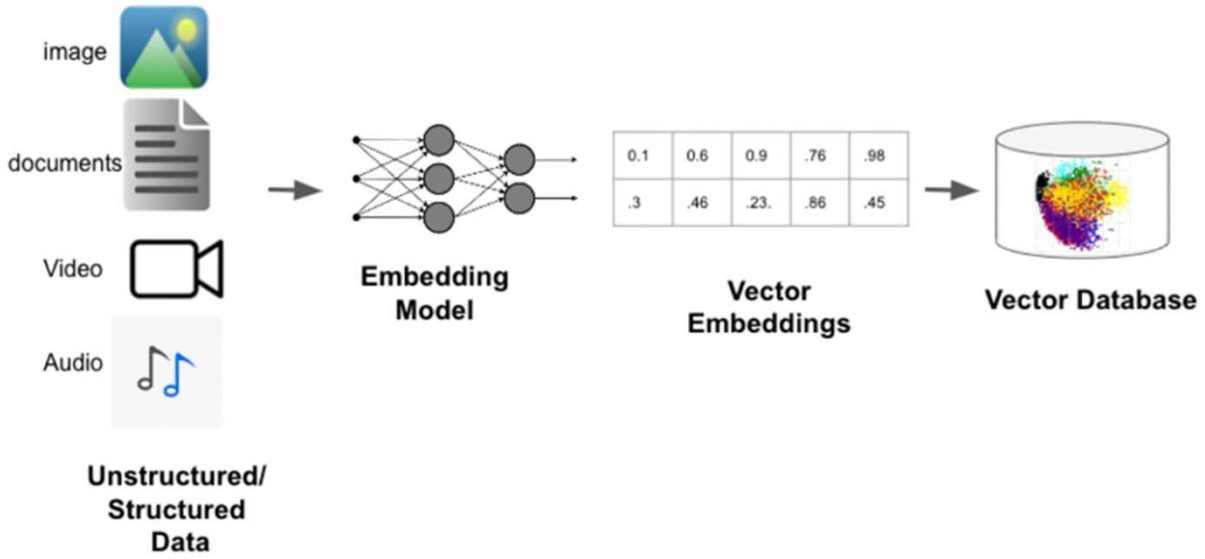


Source: Google


Vector databases
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Vector Databases

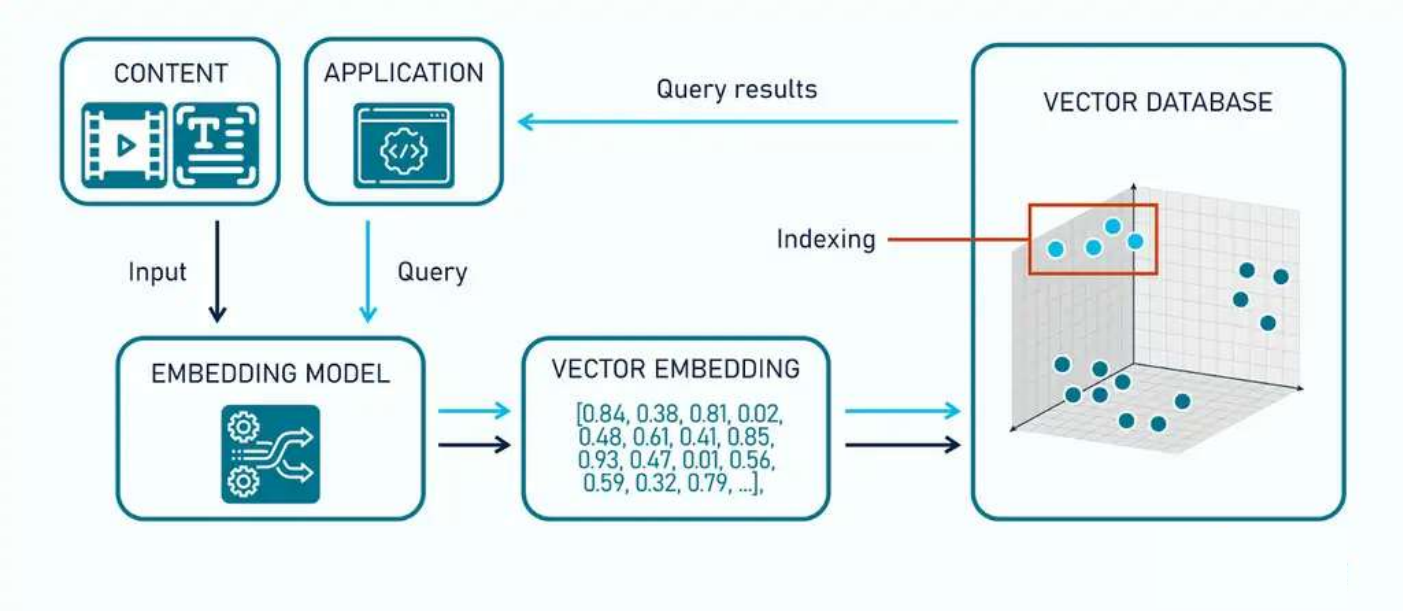
- Unlike a standard database that searches for exact keywords, a vector database searches for **similarity**



Source: IBM

Vector Database Query

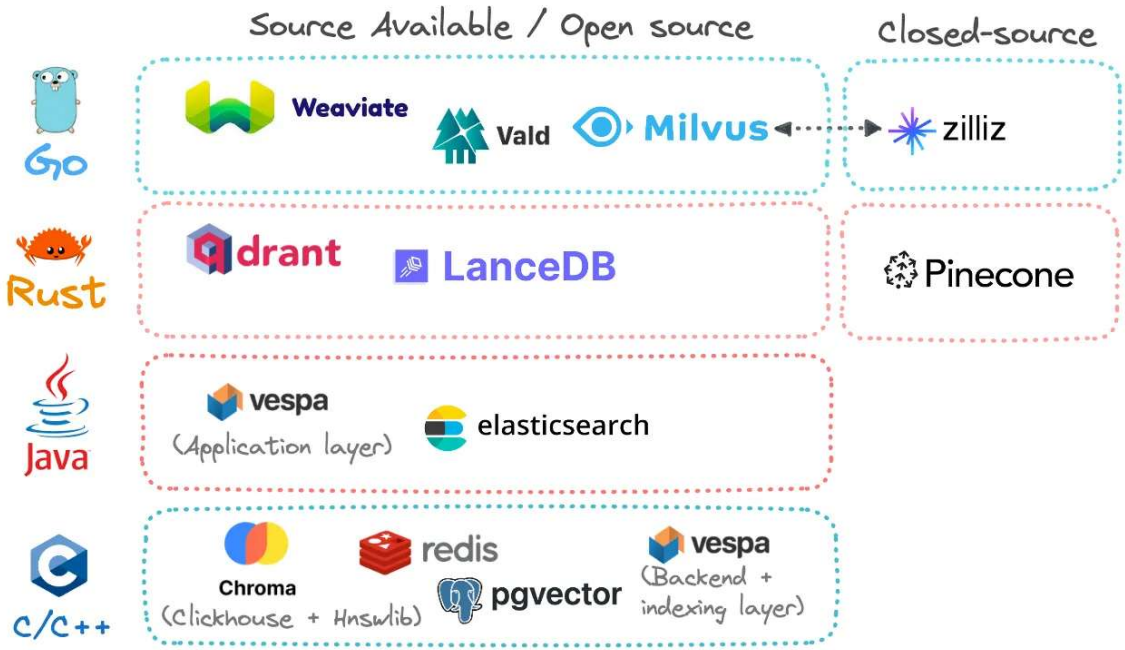
There has been a rapid growth in options



Source: altexsoft

Vector Database Landscape

There has been a rapid growth in options



Source: thedataquarry.com


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